

Methodology on the Multi-Page Visually Rich Document Understanding Frontier: A Literature Review

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Abstract

Multi-page visually rich document understanding (MP-VRDU) has emerged as a distinct research area driven by the prevalence of long-form documents such as contracts, scientific papers, and financial reports whose evidence is distributed across pages and modalities. While single-page document understanding has matured into a well-defined sub-field with established benchmarks, the multi-page setting introduces challenges that simple extension cannot address: total document length frequently exceeds backbone context windows, answer-bearing evidence is sparsely distributed, and cross-page structural relationships cannot be recovered from per-token layout signals alone. This review synthesises recent work in MP-VRDU, organising the field by how systems manage cross-page evidence rather than by OCR dependency. We examine four architectural families, three modality-specific challenges, and the inference-time and training strategies that recur across them, before identifying three persistent gaps that current approaches have not resolved: data scarcity coupled with evaluation integrity concerns, the evidence pipeline bottleneck between coverage and fidelity, and cross-page structural reasoning that remains brittle across all current routes.

1 Introduction

Visually rich documents (VRDs) such as scientific papers, business reports, financial filings, contracts, and technical manuals are pervasive in real-world workflows, and understanding them is a long-standing problem at the intersection of natural language processing, computer vision, and document artificial intelligence. Document visual question answering (DocVQA), the task of answering natural language questions over document images, has become a standard formulation for evaluating progress in this area [1]. Documents serve as the primary medium through which organisations

record and communicate information, and manually extracting specific information from them is time-consuming, expensive, and prone to human error, motivating the development of automated systems that can answer natural-language queries over document images directly. The vast majority of prior work has focused on the single-page setting, where each input is a self-contained page, and the field has matured into a well-defined sub-area with established benchmarks and a clear taxonomy of OCR-dependent and OCR-free methods [2]. Current state-of-the-art models on single-page DocVQA approach or exceed human-level performance on standard benchmarks. However, real-world documents are rarely single-page: contracts span dozens of pages, scientific papers reference figures and tables many pages apart, and technical manuals contain multi-page tables and section hierarchies that no single-page view can capture [3; 4].

Multi-page visually rich document understanding (MP-VRDU) is not a simple extension of its single-page counterpart. Three distinct challenges emerge when moving from one page to many. First, input length scales with page count, with each visually rich page contributing high-resolution figures, tables, and charts whose token cost grows linearly or worse with document size, quickly exceeding the context window of existing language and vision-language models. Second, the evidence required to answer a question is typically sparse and distributed across pages, requiring a system to navigate, retrieve, and reason across the entire document rather than read a single page in detail [5]. Capturing cross-page semantic and logical dependencies requires explicit long-range reasoning that single-page architectures were not designed to support. Third, explainability becomes more demanding, as models must not only produce an answer but also localise the pages or passages from which it was derived, a requirement that single-

page systems sidestep entirely since the input page is the answer source by construction. These pressures are compounded by the well-documented tendency of long-context models to attend poorly to evidence buried in the middle of long inputs [6], a phenomenon that exacerbates each of the three challenges above.

Early multi-page approaches adapted pre-trained language models to the document domain by treating OCR-extracted text as a long sequence, but these methods ignored visual content and discarded the layout signals that define visually rich documents. Subsequent work introduced hierarchical page-level architectures [3] and global-reasoning token mechanisms [5] to enable cross-page reasoning and interaction, achieving strong results on mid-length documents but struggling to scale to longer inputs. The past two to three years have since seen rapid growth in approaches addressing these challenges, spanning architectural extensions of single-page document transformers [3; 5; 4], adaptations of pretrained multimodal large language models (MLLMs) [7; 8], retrieval-augmented pipelines [9; 10], and increasingly agent-based systems [11; 12], often combined with task-specific training or prompted reasoning loops [13; 14]. This body of work has developed along largely parallel lines, with inconsistent terminology and limited cross-family comparison, making it difficult to identify which design choices most influence performance in the multi-page setting.

Prior surveys have charted document understanding from complementary angles: OCR-centric deep learning pipelines [15], multimodal pretraining for visually rich document content understanding [16], deep learning for information extraction over VRDs [17], and recent LLM- and MLLM-era treatments [2] including multimodal retrieval-augmented generation for long documents [18]. While valuable, these works remain anchored to the single-page setting or organise progress by OCR dependency and method-family chronology, treating multi-page documents as an incremental extension rather than a distinct problem. As a result, the concerns that define MP-VRDU receive limited systematic treatment: how architectures manage cross-page evidence under context-window constraints, how text, visual, and cross-page structural modalities trade page coverage against reading fidelity, how inference-time retrieval, reasoning, and agentic orchestration coordinate when no single forward pass can hold the document, and how vision-

language-model-synthesised data shapes both training and evaluation integrity.

This review synthesises recent MP-VRDU research with the aim of identifying persistent gaps that motivate continued research in this area. Section 2 briefly summarises the single-page mechanisms that multi-page systems inherit, supporting a generalist reader who is new to the document understanding literature. Section 3 introduces four architectural families that organise the field through the lens of cross-page evidence management. Sections 4 through 6 narrow into specific design dimensions: multimodal representation under context constraints, inference-time composition strategies, and component-level training. Section 7 catalogues the resources that support training and evaluation, and Section 8 identifies three persistent open gaps in the field.

2 Background: Single-Page VRDU

Multi-page VRDU systems inherit most of their per-page mechanisms from the single-page literature, recently surveyed by (author?) [2]. The dominant axis at the single-page level distinguishes two architectural families. **OCR-dependent frameworks** preprocess the page with off-the-shelf optical character recognition (OCR) or PDF parsers, producing recognised text and bounding-box layout that is either fed into the language model prompt directly or projected through an auxiliary multimodal encoder such as LayoutLMv3 [19], with text-layout fusion supplied through 2D positional encodings, layout-aware attention, or rule-based verbalisation of bounding boxes. **OCR-free frameworks** discard the parser and instead have a vision encoder read text directly from rendered page pixels, with the language model trained to recover text and layout through pretraining objectives such as text recognition, grounding, and captioning [20; 21]. The two families trade different costs: OCR-dependent systems inherit cumulative parser errors but avoid expensive visual pretraining, whereas OCR-free systems handle handwriting and complex layout end-to-end but require high-resolution input and large-scale pretraining.

Across both families, three orthogonal design decisions recur and persist into the multi-page setting. Text is recovered as OCR strings in the prompt, through OCR-augmented multimodal encoders, or as a training objective for the vision encoder itself. Layout is encoded as 2D positional features,

as verbalised bounding boxes in the prompt, or through supervised pretraining tasks such as text grounding. Visual tokens are managed through low-resolution patch embeddings, high-resolution shape-adaptive cropping, or compression via resamplers, Q-Formers, and convolutional reducers [8; 22]. Single-page systems make these choices assuming the entire document fits in a single backbone input; the multi-page setting reformulates each into a question of which page or region to attend to next.

3 Architectural Landscape

Where the single-page literature is naturally organised by OCR dependency, the multi-page setting introduces a different organising principle. Prior multi-page surveys have continued to use the OCR-dependency axis [2], but this axis is largely orthogonal to the central question in MP-VRDU. OCR-using systems often treat extracted text as just one component in a multimodal pipeline [11; 12], while OCR-free systems are frequently pretrained or aligned with OCR-derived supervision [10; 21]. The defining challenge in MP-VRDU is instead how a system manages document-scale evidence while bounded by context-window limits. Answer-bearing evidence is typically sparse, distributed across pages, and multimodal in nature, so models must decide what information to retain, what to discard, and how the surviving information should interact across pages. These decisions are governed by architectural design, which determines how evidence is selected, preserved, aggregated, and exposed for reasoning. We therefore organise multi-page frameworks primarily by architectural trend, with OCR dependency as a secondary descriptor.

Through this evidence-management lens, the field has shifted from encoding all pages jointly toward increasingly selective and iterative evidence handling, producing four recurring architectural trends: page-to-document encoding models, backbone-centric adaptation, retriever-generator pipelines, and agentic pipelines. Figure ?? illustrates these families, and Table ?? catalogues representative systems within each. OCR dependency remains a useful descriptor but cuts orthogonally across all four families.

3.1 Page-to-Document Encoding Models

Page-to-document encoding models encode each page through a shared document encoder including page-level OCR text, layout, and visual features, then connect page representations through a document-level exchange mechanism. These mechanisms include per-page summary tokens [3], document-global tokens [5], recurrent memory [23], and sparse cross-page attention [4]. Cross-page interaction occurs within the framework in a single forward pass, while page identity, local layout, and reading order remain explicit, providing a strong inductive bias for multi-page structure. This explicit cross-page modelling makes the family well suited to tasks where relevant evidence appears across pages within a coherent document structure, such as reports and business forms [3; 4].

The limitations of this family follow from the same design choices that give it strength. Document-level representations may lose fine-grained evidence when page information is compressed or weakly connected [5; 23], since the summary tokens or recurrent states must condense an entire page into a fixed-size representation. Computation also grows with document length, as all pages typically need to be encoded jointly in a single forward pass, making very long documents difficult to scale. As context windows of pretrained MLLMs have grown rapidly in recent years, the practical motivation for a dedicated cross-page encoder has weakened, and subsequent families have moved toward absorbing cross-page information into the backbone itself or selecting evidence before reading rather than encoding it all up front.

3.2 Backbone-Centric Adaptation

Backbone-centric adaptation removes the dedicated cross-page encoder entirely, instead processing the document inside a pretrained MLLM whose context window absorbs cross-page information directly. Adaptation strategies span a spectrum of training intensity. At the lightweight end, projectors map per-modality features into the backbone’s token space [24], adapters enable parameter-efficient fine-tuning for document understanding [?], and compression modules constrain visual-token growth on long inputs [8]. At the heavier end, the backbone itself is unfrozen for continued pretraining or instruction tuning on document corpora, trading parameter efficiency for stronger document-specific grounding [7]. These models

inherit the instruction-following and multimodal reasoning of the foundation model and remain parameter-efficient to train, fitting tasks that combine document reading with open-ended reasoning. Long-context tuning further improves MLLM performance on multi-page documents of moderate length [21; 22].

However, because the backbone must absorb document-scale evidence internally, performance degrades beyond moderate-length documents, leaving quality bound to context length and to how faithfully fine-grained text and visual signals survive token compression [25]. Reasoning over visual elements depends entirely on the backbone’s pretrained visual capabilities, and as most models in this family are OCR-free, reliable text recovery requires high-resolution input that aggressive down-sampling readily compromises. Document structure such as section hierarchy and figure-caption linkage must likewise be recovered implicitly from pixels rather than supplied through layout tokens or a structural encoder [26; 7]. These limits motivated moving evidence selection out of the backbone altogether.

3.3 Retriever-Generator Pipelines

Retriever-generator pipelines divide MP-VRDU into evidence selection and answer generation, where a retrieval module returns a compact subset of OCR chunks, pages, visual patches, multimodal elements, or graph nodes, and a generator answers conditioned only on those units [9; 10]. The design follows the broader retrieval-augmented generation paradigm [27; 28] adapted to visually rich inputs. Inference cost depends on the retrieved content rather than document length, allowing this architecture to scale to documents that exceed any current MLLM context window. Modular decomposition allows the retriever and generator to be swapped independently, while retrieved evidence provides a partial audit trail for the answer. The same interface extends from a single document to a corpus, supporting multi-document question answering without significant architectural change. Structured domains with reliable retrieval anchors such as scientific papers and technical manuals benefit most from this design [29], and visually rich pages benefit further when page-level visual retrieval preserves figures, tables, and layout that OCR-only pipelines lose [10; 30].

The limitations of this family stem from its single-pass control flow. Evidence missed at re-

trieval is unrecoverable at generation, and broadening the retrieved set to compensate introduces noise and attention dilution. Performance depends heavily on parser and embedding quality, while chunking errors, OCR mistakes, and retriever-generator modality mismatches propagate directly into the answer. The single-pass design also presupposes that relevant evidence is rankable by query similarity, which weakens when answers require aggregating pages that are not individually strongly relevant [31; 32]. These pressures motivate replacing the fixed pipeline with adaptive, multi-step reasoning.

3.4 Agentic Pipelines

Recent surveys characterise agentic LLM systems through their capacity to reason, act, and interact with an external environment beyond a single forward pass [33; 34]. Following this framing, we categorise MP-VRDU systems that perform multi-pass evidence acquisition or distribute work across specialised agents as agentic, while reserving the retriever-generator family for single-pass models. Agentic systems treat MP-VRDU as an iterative evidence-seeking process, where one or more controller MLLMs decide what to inspect next from intermediate observations and issue tool calls for retrieval, navigation, cropping, or OCR [35; 11]. Roles such as retrieval, inspection, synthesis, and adjudication may be distributed across agents, with each agent backed by a different backbone model selected for its subtask [14] rather than tied to a single generator as in retriever-generator systems. This design lifts the single-pass bottleneck, allows compute to scale with question difficulty rather than document length, and renders the trajectory itself an inspectable audit trail. The family is well suited to questions whose evidence is distributed across pages or modalities and to multi-hop questions that benefit from query reformulation between steps [12; 11], as well as visually heterogeneous documents whose text, tables, and figures benefit from tool-specific inspection [14; 36].

The defining property of agentic pipelines is that the trajectory adapts to question difficulty rather than following a fixed control flow; multi-component systems with predetermined execution order, even when described as “agents”, remain within the retriever-generator family. This adaptability comes at several costs. Inference cost is difficult to bound before execution, as the variable-length action sequence can extend arbitrarily for hard questions. Action policies and selection cri-

teria remain sensitive to prompt phrasing, with silent failure when the controller commits early to wrong evidence. Reproducibility is complicated by stochastic generation and the absence of a single attributable forward pass. Finally, the system inherits the recall ceiling of every retriever, parser, or OCR module it calls, so weaknesses in any tool propagate into the final answer.

4 Multi-Page Multimodal Representation

While Section 3 categorises MP-VRDU systems by where cross-page evidence is processed, this section analyses the problem from a modality perspective. Single-page VRDU already handles text, vision, and layout through the well-established mechanisms outlined in Section 2. The multi-page setting introduces a shared pressure across all three modalities: the full document often exceeds the backbone’s context window, and even when it fits, the answer occupies only a small fraction of a long input where attention is diluted and key evidence is easily lost in the middle [6]. This forces each modality to trade page coverage against reading fidelity, but the resolution of this trade-off differs across text, vision, and structure.

4.1 Text Modality

Text is the most information-dense modality in MP-VRDU, carrying the bulk of a document’s explicit semantic content in a compact, searchable form, obtained either through OCR or recovered directly from page pixels by a vision-language model. Extracted OCR text supports cheap indexing and retrieval over documents that exceed context windows, natural answer-passage attribution, and bridges to visual elements via captions and references. However, this comes at the cost of OCR error propagation, fragile multi-column and table reading order, and the loss of chart, layout, and handwriting semantics that no text extraction can capture. Pixel-recovered text avoids OCR dependence but demands high-resolution input and extensive text-recognition pretraining, both of which scale poorly with document length.

Three patterns have emerged for handling text under these challenges. **Per-page aggregation** encodes each page in detail and combines per-page representations into compact cross-page summaries through attention or memory tokens, as in Hi-VT5 [3], GRAM [5], and Arctic-TILT [4]. The page-to-document transformer family typically follows this

pattern, treating OCR-extracted text as the primary per-page input and encoding it jointly with layout and visual embeddings before document-level aggregation. **Whole-document compression** retains all pages within a single backbone context by aggressively reducing per-page text tokens through resamplers, layout-aware compressors, or token-reduction modules, as in mPLUG-DocOwl2 [8] and TextHawk2 [22]. Backbone-centric MLLMs typically follow this pattern, reading text through the backbone’s internal vision capabilities applied to whole-document context. **Retrieve-then-read** abandons full-document encoding entirely, indexing text at chunk or page level and passing only a retrieved subset to the reader, as in CREAM [37], MultiDocFusion [29], DocAgent [11], and MDocAgent [14]. Agentic variants extend this into multiple selection-reading rounds, using text not only as a reading target but as a navigation layer where outlines and section searches direct later visual inspection.

Each pattern carries a characteristic risk: per-page aggregation discards token-level detail in the summary step, whole-document compression sacrifices fine-grained reading on individual pages, and retrieve-then-read makes evidence missed at retrieval unrecoverable at the reading stage. The latter two routes also depend on document-level structure to decide which pages or sections to read in detail, an issue we return to in Section 4.3.

4.2 Visual Modality

Visual elements in MP-VRDU range from text-bearing structures such as dense paragraphs and tables to non-textual evidence such as charts, figures, and handwriting that OCR cannot recover. Visual token cost scales jointly with page count and per-page resolution, making full-document high-resolution encoding infeasible at realistic document length. This forces every architecture to trade page coverage against reading fidelity within a fixed token budget, and the resolution of this trade-off varies more sharply than in the text case because the floor below which visual evidence becomes unreadable is harder to predict from the question alone.

Three patterns have emerged for managing this tension. **Aggressive token compression** keeps every page available in context but reduces per-page visual tokens through resamplers, Q-Formers, or convolutional reducers, as in mPLUG-DocOwl2 [8] and TextHawk2 [22]. This preserves cover-

age at the cost of per-page detail and works best for documents where coarse visual cues such as section headings and large figures dominate the answer-bearing content. **Selective high-resolution reading** restricts expensive encoding to a small subset of pages surfaced by retrieval, navigation, or controller-driven inspection, sending those high-fidelity tokens into the reader while leaving other pages unread, as in Doc-V* [35] and SimpleDoc [38]. Many systems pair low-resolution thumbnails for navigation with selective high-resolution inspection of pages a retriever or tool call surfaces. **Retrieval-only visual embeddings** decouple encoding from reading altogether: page images are encoded purely to produce retrieval embeddings, and detailed reading of selected pages is handed to a separate text or vision module, as in M3DocRAG [9] and VDocRAG [10].

Each pattern carries a characteristic failure mode. Aggressive compression loses fine text and small visual elements, with chart axis labels, table cell values, and footnote text the first casualties. Selective high-resolution reading inherits the recall ceiling of whichever tool selects the pages, since a fine pass cannot recover when the coarse pass misroutes. Retrieval-only visual embeddings may rank layout-similar pages without confirming that the answer-bearing region is actually present, particularly when the question asks for content that is visually similar but semantically distinct from the surrounding pages. Compression and tokenisation choices further determine how reliably embedded text, table cells, and figure annotations survive into the model input.

4.3 Cross-Page Structure

Layout in MP-VRDU spans two levels. Within a page, it captures reading order, columns, table structure, and bounding-box geometry, largely shared with single-page VRDU and covered in prior surveys [2]. Across pages, it captures document-level structure such as section hierarchies that span multiple pages, page order, multi-page tables, and cross-references between figures, captions, and the text that describes them; this is the distinctive multi-page problem. Within-page layout is largely handled by mechanisms inherited from single-page VRDU, either through layout-aware pretraining objectives [3; 8?] or through implicit recovery from page pixels by an OCR-free MLLM [22]. Cross-page layout is rarely captured by these mechanisms, as per-token spatial annotations de-

scribe individual tokens rather than relationships between sections, pages, or references, and pre-training supervision for document-scale structure is expensive to obtain.

A growing line of work therefore exposes parsed structural information directly to the model, and three patterns have emerged. **Structure-aware chunking** uses a parser to extract section trees and heading hierarchies, then segments documents along semantic boundaries so retrieved units respect section scope rather than sliding-window cuts, as in MultiDocFusion [29]. This approach preserves parent context for each retrieved chunk and avoids the common failure mode of fixed-window chunking, which can split a section header from its content or cut across table boundaries. **Hierarchical retrieval indices** route queries first to coarse units such as sections or chapters and then to fine-grained chunks within the selected branch, reducing irrelevant pages while preserving cross-page context, as in MHier-RAG [39] and MLDocRAG [40]. The hierarchical decomposition addresses the cross-page fragmentation that single-tree designs leave unresolved. **Graph-based traversal** converts headings, entities, and cross-references into a knowledge graph, allowing controllers to follow typed edges across pages rather than relying on dense similarity alone, as in KGP [31]. Graphs can be constructed at index time over the full document or per query, with the latter allowing the structure to adapt to the question.

Each route carries a distinct cost. Layout-as-objective demands pretraining supervision that is expensive at multi-page scale, since cross-page structural annotation does not transfer cleanly from single-page corpora. Implicit visual layout depends on backbone capability and high-resolution input that aggressive downsampling readily compromises. Explicit parsed structure depends on parser quality, with errors in heading detection, table boundaries, and reading-order extraction propagating into every downstream retrieval and navigation step.

5 Inference-Time Adaptation

Inference-time adaptation refers to strategies that improve MP-VRDU performance through changes to the inference pipeline rather than to model parameters. A frozen language or vision-language backbone is paired with off-the-shelf retrievers, structural parsers, prompted reasoning loops, or

multi-agent orchestrations, none of which require gradient updates on the backbone itself. This approach is particularly suited to MP-VRDU settings where supervised data is scarce and expensive to collect, where backbones must remain exchangeable across systems, or where rapid adaptation to new document types is needed without retraining. The data scarcity issue is especially acute in MP-VRDU, as we discuss in Section 8, which has contributed to the growing prevalence of training-free strategies in agentic and retrieval-based systems. We organise these strategies along the inference pipeline: evidence selection through retrieval and navigation (Section 5.1), reasoning over the assembled evidence (Section 5.2), and coordination across multiple frozen components (Section 5.3).

5.1 Retrieval and Navigation

Inference-time retrieval selects a compact subset of pages, chunks, or graph nodes for the reader using off-the-shelf encoders, structural parsers, or LLM-based rerankers, without fine-tuning the retriever on document data. Retrieval quality bounds every downstream reasoning step, so the choice of signal dominates system behaviour on long documents. Two broad families have emerged, distinguished by whether passages are scored in isolation or navigated through explicit relations.

Similarity-based retrieval. Similarity-based methods score each passage against the query independently, using sparse lexical signals, dense embeddings, or combinations of both. Sparse signals such as BM25 match on lexical or structural overlap and miss semantic connections and paraphrased queries, resulting in degraded performance on questions whose phrasing diverges from the document’s surface form. Dense embedding retrieval addresses this by encoding pages or chunks into learned representations so queries match on semantic similarity rather than lexical overlap. The dominant dense signal in MP-VRDU is vision-language late-interaction encoding such as ColPali [41] or its Qwen-based variants, which extends the late-interaction paradigm of ColBERT [42] to page images. These encoders embed the query as a sequence of contextual token vectors and each page as a bag of patch embeddings produced by a vision-language transformer over the rendered page image, then score a query-page pair by taking the maximum cosine similarity of each query token over all page patches and

summing the result. This maximum-similarity aggregation preserves fine-grained alignment between specific query terms and specific page regions, which is critical in MP-VRDU because a question token may align with a single figure caption, table cell, or paragraph buried in a long document.

M3DocRAG [9] instantiates this pattern by retrieving at chunk and page granularity with the rest of the pipeline frozen. AVIR [43] adds an adaptive page selector that examines the per-query score distribution and either thresholds at a fixed cutoff for short documents or applies clustering with a top- K bound on longer ones, so the number of pages handed to the reasoner scales with the score distribution rather than being fixed. Joint variants combine sparse and dense signals at the reranking stage: CREAM [37] pairs dense coarse retrieval with iterative LLM-based reranking across overlapping chunk groups, so dense semantic candidates are filtered by an LLM judgement that draws on the full passage text. SimpleDoc [38] combines ColPali page embeddings with LLM-generated summary reranking under a working memory that persists across rounds, so successive iterations can refine the joint score. MDocAgent [14] runs parallel text and image retrieval tracks whose candidates are aggregated by a downstream agent, joining textual and visual evidence rather than fusing modalities at encoding time. Such combinations improve recall by spanning what any single signal misses, but still score each candidate passage in isolation, so retrieval cannot follow logical links, cross-references, or multi-hop dependencies that span passages.

Structure-based retrieval. Structure-based methods exploit a parsed representation of the document to make passage relations first-class objects, addressing the structure-blindness of independent scoring. Tree-based variants such as MultiDocFusion [29] use a hierarchy parser to recover document, section, and subsection trees from scanned PDFs and group passages along depth-first-search-ordered Markdown headers, so retrieved chunks preserve their parent context. DocAgent [11] extracts a tree-formatted XML outline from headings, sections, and captions and exposes it through section-level search and section-content fetch tools that navigate the parsed hierarchy directly. MHier-RAG [39] extends the tree structure to a dual hierarchy of in-page flattened chunks and cross-page topological

chunks, with LLM-based reranking across both indices to address cross-page fragmentation that single-tree designs leave unresolved.

Graph-based variants treat headings, entities, and cross-references as graph nodes with typed edges, allowing controllers to traverse from one passage to related ones rather than relying on similarity alone. KGP [31] constructs a knowledge graph over passages and document-structure nodes such as pages, paragraphs, and tables, with edges encoding lexical similarity and structural belonging, then uses an instruction-tuned LLM as a traverser that predicts the next evidence node from the visited path. MoLoRAG [32] builds a page graph in which edge candidates are filtered by embedding similarity, then scores logical relevance with a vision-language prompt and combines that score with semantic similarity for multi-hop traversal, so retrieval can follow logical dependencies that surface similarity misses. MLDocRAG [40] constructs the graph per query, anchoring chunk nodes to vision-language-model-generated query nodes and expanding through nearest-neighbour edges between queries, so the structure adapts to the question rather than being fixed at index time. Structure-based methods recover references that similarity scoring misses, at the cost of construction overhead, edge-quality dependence, and traversal cost that grows with hop count.

5.2 Inference-Time Reasoning Strategies

Inference-time reasoning strategies govern how a frozen language or vision-language model accesses, updates, and reasons over evidence during decoding, without modifying model parameters. In MP-VRDU, these strategies can be broadly categorised by whether evidence acquisition remains static or becomes dynamically interactive during inference. **Closed-buffer reasoning** operates over a fixed evidence set assembled before generation begins, whereas **tool-interactive reasoning** allows the model to iteratively acquire additional evidence through retrieval or page-level actions during the reasoning process.

Closed-buffer reasoning. Closed-buffer reasoning performs multi-step reasoning over a fixed evidence buffer assembled by a prior retrieval step or by a compressed document representation. Chain-of-thought prompting [44] is the canonical instantiation, eliciting intermediate reasoning steps that integrate retrieved text, tables, and figures and com-

bine facts across pages before producing the final answer. The entire reasoning chain runs over the assembled buffer, so the reasoner cannot acquire new or finer-grained evidence once decoding begins. MHier-RAG [39] applies a structured chain-of-thought prompt to its hierarchical retrieval output, although a single reasoning chain remains vulnerable to compounding errors on dense layouts. SimpleDoc [38] interleaves self-refinement and self-verification, emitting a refined query when evidence is judged insufficient, although the loop terminates on the backbone’s own judgement and inherits its blind spots. Across these variants, reasoning operates over a fixed evidence buffer, so questions whose evidence sits outside the retrieved set remain unreachable regardless of how the trace is sampled or verified.

Tool-interactive reasoning. Tool-interactive reasoning interleaves reasoning and evidence acquisition during inference, allowing newly observed pages to guide subsequent reasoning steps. The ReAct paradigm [45] is the canonical instantiation, alternating intermediate reasoning traces with tool calls whose observations feed the next step. In MP-VRDU the tools retrieve, fetch, or inspect document pages, letting the reasoner gather evidence during inference rather than work from a closed buffer. Doc-React [46] instantiates flat-retrieval ReAct with sub-query refinement and entropy-based termination, although each iteration still selects from a single retriever’s ranked list and recovery from a poor encoder match relies on subsequent query rewrites. Outline-driven variants combine this loop with the structured navigation of Section 5.1: DocAgent [11] traverses its tree-formatted XML outline through section-level search and fetch tools chosen turn by turn by an actor agent, inheriting the parser’s outline quality and degrading on documents without exploitable structure.

A related pattern is coarse-to-fine visual reading, which addresses the fixed-resolution limit of single-pass reading by inspecting retrieved pages selectively at multiple scales rather than as whole images. Doc-V* [35] maintains a thumbnail-level overview and applies high-resolution reading only on regions returned by retrieval or fetch actions, while DocLens [36] and ViDoRAG [12] apply the same split through multi-agent pipelines, with DocLens separating a page navigator from an element localiser that crops layout regions before reasoning. Such designs concentrate compute on task-relevant

regions, yet make reasoning conditional on the localiser, since the fine pass cannot recover when the coarse pass misroutes. Across the tool-interactive family, adaptive evidence acquisition improves localisation and supports more exploratory multi-page reasoning, but introduces tool latency that grows with rounds, sensitivity to prompt design and to early action choices that subsequent steps cannot recover from, and attention dilution within fetched pages at their fixed reading resolution.

5.3 Agentic Orchestration

Retriever-generator and agentic pipelines often distribute a single MP-VRDU question across multiple specialised language or vision-language model calls, each handling a particular modality, reasoning step, or verification role. The frozen backbones are not modified; coordination is achieved through prompt design and call ordering. The two families correspond directly to whether the trajectory of calls is fixed at design time or adapts to the question at inference: retriever-generator pipelines run a trajectory fixed in advance regardless of how many agents they invoke, while agentic pipelines treat the call sequence itself as a variable.

Fixed-trajectory pipelines. Fixed-trajectory pipelines follow a control flow set in advance: which calls are made, in what order, and how their outputs are combined are all determined at design time. Modality-partitioned variants dispatch readers by input type and aggregate their outputs through a final synthesis step. MDocAgent [14] runs a five-agent pipeline in which a general agent provides initial multimodal context, a critical agent extracts key evidence from both modalities, dedicated text and image agents read their respective tracks in parallel, and a summariser fuses the four outputs into a single answer. M2RAG [47] splits work between a text filter, a visual extractor, and a modal fuser that applies a confidence-conditional rule, falling back to one modality when the other reports no information and fusing only when both contribute. These parallel pipelines improve modality coverage compared to single-track retrieval, but lack an error-correction stage, so an incorrect output from one branch can dominate aggregation when the other branch lacks counter-evidence. A wrong choice of routing or ordering cannot be revised mid-execution.

Adaptive-trajectory pipelines. Adaptive-trajectory pipelines treat the call sequence itself as a variable: which agents are invoked and how many rounds the loop runs are decided per question rather than fixed in advance. Loop-based variants such as ViDoRAG [12] cycle a seeker that retrieves coarse candidates, an inspector that reviews them at fine resolution and emits reflective feedback, and an answerer that performs a consistency check, with a Gaussian mixture model setting per-query bounds for each modality from the similarity distribution rather than using a static top- K . Proposer-verifier variants insert an explicit verification stage between candidate generation and final answer. DocLens [36] samples multiple candidate answers from a reasoning module and routes them through an adjudicator that compares reasoning paths and selects the most consistent conclusion, so candidates are judged against each other rather than fused unconditionally. DocAgent [11] pairs an actor with a reviewer that cross-checks through complementary tools, with low tool-overlap reported between the two agents, and writes any disagreement into a shared reflection memory that updates the actor’s behaviour on subsequent rounds. Adjudication catches single-branch errors that fusion cannot, but the verifier draws on the same backbone family as the actor, so shared blind spots persist and the verifier can over-reject correct candidates under noisy tool output. Adaptive-trajectory pipelines improve the accuracy-latency trade-off, but inference cost remains hard to bound before execution, scaling jointly with the number of calls involved.

6 Training Strategies

While Section 5 held the backbone fixed, many MP-VRDU systems extend these pipelines by training specific components. We organise MP-VRDU training by the inference-time component being trained, namely the retriever, the reasoner, and the agentic orchestration policy. Two cross-cutting techniques, parameter-efficient tuning and multi-stage curriculum training, recur across all three targets.

6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali [41] score pages in isolation, leaving three gaps that retriever training aims to address: alignment between visual

document embeddings and the text they should match, fusion of evidence that is spread across multiple pages, and the modelling of logical relevance beyond surface similarity. Modality-alignment variants pretrain a visual document encoder against OCR text. VDocRAG [10] uses self-supervised representation compression and generation objectives that align visual document representations with OCR-derived textual content, enabling vision-language models to learn document embeddings directly from unlabelled document-image pairs and compressing page images into single-vector embeddings that retain text alignment without late-interaction cost. Cross-page-fusion variants freeze the encoder and learn a neighbour-blending head. DFVC [48] adds a multi-layer perceptron adapter with gating and residual connections that fuses each page embedding with its neighbours, so evidence spread across pages becomes retrievable as a coherent unit rather than scored independently. Logical-relevance variants train a small scorer over a page graph. MoLoRAG [32] fine-tunes a vision-language scorer that ranks neighbour pages during graph traversal, supplying the cross-page logical signal that embedding similarity alone often misses.

6.2 Trained Reasoning Components

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating approaches that train the reasoning component directly. Reward-driven variants train a reasoning policy with explicit evidence supervision. DocR1 [13] fine-tunes a vision-language backbone with evidence-page-guided reinforcement learning, where the reward combines format compliance, answer accuracy, and a page-citation indicator. The training procedure encourages a coarse-to-fine reasoning strategy where the model first identifies relevant evidence pages before generating answers, producing structured outputs that interleave thinking, evidence pages, and answers. Trace-imitation variants instil a reasoning chain through supervised fine-tuning and preference optimisation. Chain-of-Reading [49] bakes in a four-stage plan-execute-integrate-verify chain through staged supervised tuning followed by direct preference optimisation on curated preference pairs, internalising a multi-step reasoning trajectory rather than relying on prompting alone. Uncertainty-driven variants train an aggregation head rather than a trajectory. DocSLM [26] learns an entropy-based calibrator that se-

lects the lowest-uncertainty answer across streaming segments, providing a self-consistency signal under tight parameter budgets that trades compute for robustness against single-pass errors.

6.3 Trained Agentic Orchestration

Prompt-only orchestration is unreliable on smaller open-weight backbones that cannot follow ReAct protocols, motivating training approaches that internalise tool-use behaviour. Behavioural-cloning variants distil agent trajectories from a stronger model. DocDancer [50] fine-tunes an agent on synthesised exploration-then-synthesis traces that pair retrieve and read actions with intent-conditioned tool selection, so action choices are conditioned on the agent’s evolving goal. Reinforcement-learned variants reward correct tool use and trajectory shape. Doc-V* [35] couples supervised fine-tuning on coarse-to-fine ReAct traces with reinforcement learning over a two-action space of retrieval and fetch calls plus entropy-based stopping, training the agent to balance exploration of new pages against termination on sufficient evidence. MACT [51] decomposes the loop further into planning, execution, judgement, and answer agents, separating the judge from the corrector to avoid the same agent excusing its own errors, and applies agent-wise adaptive test-time scaling so compute is allocated per role and per question based on difficulty.

6.4 Cross-Cutting Training Techniques

Two techniques recur across retriever, reasoner, and orchestrator training. Full backbone fine-tuning is prohibitive at long-document context lengths and erases pretrained capabilities, so the dominant pattern reuses strong pretrained LLMs or vision-language models and updates only a small number of task-specific components. Rather than training vision encoders or language backbones end-to-end, these methods freeze most pretrained parameters and introduce lightweight modules such as LoRA adapters [52], projection heads, layout tokenisers, or task-specific heads. LayTokenLLM [?] exemplifies this by keeping the language model backbone frozen and introducing a lightweight layout tokenisation module together with LoRA-based adaptation. Rather than modifying the architecture, it injects layout information into the input sequence using a compressed representation and a specialised positional encoding scheme that avoids additional context overhead, enabling efficient scaling to long multi-page documents. The

same pattern appears in DFVC’s frozen-ColPali adapter, DocDancer’s LoRA trajectory tuning, and CREAM’s adapter-based answerer, confirming that parameter-efficient tuning versus full fine-tuning is more often a compute-budget decision than a target-component one.

Separately, several systems converge on multi-stage curricula that start on single-page or local objectives before introducing document-scale supervision. Directly adapting single-page models to multi-page objectives often fails to capture cross-page dependencies and long-context interactions, so progressive training extends models from simpler local tasks to more complex multi-page objectives. mPLUG-DocOwl2 [8] uses three stages of single-image structure learning to ensure that compressed visual tokens preserve document text and layout information, followed by multi-page parsing pretraining and multi-task fine-tuning. DocSLM [26] extends this with a five-stage schedule beginning with image-OCR alignment, then image-level and document-level compression, then instruction tuning, and finally streaming-abstention calibration that teaches the model to decline answering when its uncertainty is high.

7 Datasets

Datasets in MP-VRDU span three roles: pretraining, fine-tuning, and benchmarking. Pretraining historically relied on single-page archives such as IIT-CDIP [53] and OCR-IDL [54], with genuinely multi-page corpora only recently emerging through DocStruct4M’s mixed single- and multi-page pretraining [21] and the purely multi-page MP-DocStruct1M, Doc-750K, and scientific PaperPDF [8; 7; 30]. Three pretraining recipes recur. The first is OCR and layout self-supervision over scanned archives such as IIT-CDIP or CCpdf [55]. The second is document-structure pretraining over synthesised question-answer pairs, supplying weakly-supervised text-to-position alignment through corpora such as DocStruct4M, the purely multi-page MP-DocStruct1M, and the scientific PaperPDF. The third is generic vision-language pretraining followed by document continue-pretraining, in which a web-scale vision-language corpus such as Cauldron [56] is combined with document-specific data. The field is converging on the third recipe, with token-level structure prediction over synthesised question-answer data displacing simpler OCR-text masked-language-

modelling as the dominant document pretext task.

Fine-tuning has shifted toward bespoke synthetic generation per model, with strong vision-language models writing question-answer or trajectory data over scraped PDFs since per-page annotation does not scale to long documents [24; 38; 13; 50]. Two strategies dominate. Task-mixture fine-tuning aggregates a broad pool of public sources such as DocVQA [1], ChartQA [57], InfographicVQA [58], DUDE [?], MP-DocVQA [3], and TAT-DQA [59] into a unified multi-task mix. Capability-targeted fine-tuning instead curates a single corpus, typically synthesised by a strong vision-language model, such as the trajectory data used for agentic training [50] or reasoning-trace data used for chain-of-reading training [49]. Reinforcement and preference learning is emerging at small scale through evidence-page-guided reinforcement learning [13] and preference-pair training [49]. A defining feature of this stage is heavy reuse of evaluation datasets for training, with the same documents appearing in fine-tuning mixes and evaluation suites of the same systems, an issue we discuss in Section 8.

Benchmarks have evolved from mid-length extractive evaluation on MP-DocVQA, DUDE, and SlideVQA [3? ; 60] to long-context multimodal evaluation on MMLongBench-Doc [61], LongDocURL [62], and DocBench [?], with retrieval and page-localisation increasingly scored alongside answer accuracy [63; 13; 36]. The mid-length default suite is dominated by MP-DocVQA (5,928 documents averaging 8.3 pages with 46K questions), DUDE (5,019 documents averaging 5.7 pages with 41.5K questions), and SlideVQA (2,619 decks of approximately 20 slides with 14.5K questions). The long-document frontier is dominated by MMLongBench-Doc (135 documents averaging 47.5 pages with 1,082 questions), often abbreviated “MMLongBench” in body text yet distinct from the unrelated benchmark of the same shorthand [?]. LongDocURL (averaging 85.6 pages per document) and DocBench act as close peers, and M-LongDoc [64] pushes length further. Domain-specific evaluation through CUAD [65], TAT-DQA, and BRIDGE [66], and open-domain retrieval evaluation through M3DocVQA [67], ViDoSeek [12], and OpenDocVQA [10], complement the general-purpose suites. Evidence-aware evaluation is growing through UDA’s PaperTab and FetaTab sub-tasks [63] and the evidence-page metrics of MMLongBench-Doc, LongDocURL,

DocR1, and DocLens. Five benchmarks (MP-DocVQA, MMLongBench-Doc, DUDE, DocVQA, and InfographicVQA) appear in roughly half or more of recent MP-VRDU papers and constitute the de-facto axes for cross-model comparison. A detailed catalogue with per-dataset statistics is provided in Table ??.

8 Open Gaps

The preceding sections show that MP-VRDU has produced a rich landscape of architectures, modality-handling strategies, inference-time compositions, and training recipes. Despite this rapid progress, three persistent gaps cut across the four architectural families and remain unresolved by any current approach. We outline each in turn.

8.1 Data Scarcity and Evaluation Integrity

Acquiring high-quality multi-page annotation is impractical at scale, as per-page labelling does not extend to documents averaging eight to twenty pages and exceeding fifty pages in long-document benchmarks. Most systems therefore rely on vision-language-model-synthesised question-answer pairs and trajectories generated over scraped PDFs [24; 38; 13; 50]. These synthesis pipelines often lack independent validation, and the resulting bespoke datasets are rarely released publicly, limiting reproducibility and cross-system comparison.

The shift toward synthesised training data has compromised evaluation integrity in a structural way. The majority of papers that introduce a synthesised corpus source their PDFs from existing evaluation benchmarks, and MP-DocVQA alone appears in fine-tuning mixes and evaluation suites of roughly half the recent MP-VRDU literature, with DUDE and DocVQA showing similar overlap. The result is a contamination loop in which a model trains on a strong vision-language model’s behaviour over benchmark PDFs and is then evaluated against those same PDFs under metrics often judged by the same model family, producing a circular evaluation regime in which performance reported as zero-shot may include the test corpus in its training history. Few papers in this corpus report human-rated synthetic-data quality audits, document-hash de-duplication against benchmark test pools, or backbone-level contamination checks. Agentic stochasticity further complicates cross-system comparison, as multi-pass pipelines with prompt sensitivity and non-deterministic genera-

tion produce results that are difficult to reproduce without the original prompts and seeds. Per-paper benchmark curations, where systems are evaluated on bespoke subsets of standard suites, block direct comparison even when the same nominal benchmark is reported.

Multilingual coverage is a further blind spot, as dominant benchmarks remain overwhelmingly English-centric, with non-English layouts, reading orders, and document conventions underrepresented in both training and evaluation. These limitations leave open the design of held-out closed-test benchmarks with mandatory contamination declarations covering pretraining, fine-tuning, and backbone leakage, generator-evaluator separation when synthetic data is involved, trajectory-level metrics paired with retrieval and grounding scores, and multilingual corpora that cover non-English layouts and conventions.

8.2 The Evidence Pipeline Bottleneck

In practice, MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM’s compute and context budget [61]. Every architectural family addresses this through some lossy compromise that silently drops fine-grained content such as small text, table cells, and figure annotations. Compression is typically fixed at index time and independent of the question, so evidence may be discarded long before reasoning begins, and the model cannot recover content the encoder did not preserve. Retrieval bounds the upper limit of answer accuracy, while broadening the retrieved set introduces noise and attention dilution that can degrade reasoning even when the right evidence is present. Existing retrieval approaches often exhibit brittle behaviour through chunking errors, OCR and parser propagation, modality mismatch between retriever and reader, and per-passage scoring that ignores cross-reference and logical links. Iterative retrieval lifts this ceiling partially, yet inherits the recall of every tool or module it calls, so a weakness in any single component constrains the entire pipeline.

The bottleneck is fundamental in the sense that across page-to-document, backbone-centric, retriever-generator, and agentic families, no architecture has decoupled answer quality from a coverage-versus-fidelity trade-off. Page-to-document encoders lose fine-grained evidence during summary aggregation; backbone-centric mod-

els suffer attention dilution and the well-known lost-in-the-middle problem on long contexts; retriever-generator pipelines are bounded by retrieval performance; agentic systems compound this by also depending on tool reliability and adding orchestration latency. The trade-off is shifted between families rather than removed. These limitations leave open query-conditional and modality-aware compression that retains evidence dynamically based on the question’s evidential demands, and retrieval that follows cross-references and logical relations rather than scoring passages in isolation.

8.3 Cross-Page Structural Reasoning

Within-page layout is largely inherited from single-page VRDU through 2D positional embeddings and spatial pretraining (Section 2), while cross-page structure such as section hierarchies, page order, multi-page tables, and inter-page references remains the distinctive multi-page problem. Despite progress in this area, no MP-VRDU model performs robustly on cross-page reasoning, and document-level structure remains brittle across all three current routes.

Layout-as-objective demands pretraining supervision that is expensive at document scale, since reliable cross-page structural annotation does not transfer cleanly from single-page corpora and requires either manual labelling that does not scale or synthetic generation that inherits its own quality concerns. Implicit pixel-based recovery depends on backbone capability and high-resolution input that aggressive downsampling readily compromises; the same compression that makes long-document encoding feasible erodes the structural cues a backbone would need to recover them implicitly. Explicit parsed structure inherits parser fragility on headings, table boundaries, and reading order, with errors propagating into every downstream retrieval and navigation step. Section detection on heterogeneous PDFs is itself an open problem, and downstream methods that assume clean section trees fail silently when the parser produces inconsistent or incomplete structure.

The three routes are not exclusive, and some systems combine them, but no current approach addresses the structural problem end-to-end without falling back on parsers tuned for single-page or template-bound documents. These limitations leave open cheaper structural supervision, learned cross-page representations that survive compression, and end-to-end structure-aware retrieval that

does not depend on offline parser output.

9 LLM Usage and Other Notes

LLM Usage. Large language models, specifically Claude Opus 4.7 and GPT 5.5, were used throughout the preparation of this review for spelling and grammar correction, sentence-level rewording, reading-flow refinement, and other writing-quality tasks. All conceptual content, argumentation, and selection of cited works are my own. The LLM was used as a writing aid rather than as a source of claims, and all factual statements, citations, and quantitative figures have been verified against primary sources before inclusion. Intellectual responsibility for the final document rests with the author.

Relationship to a Companion Survey Paper.

Sections 3–?? were adapted from a survey paper on the MP-VRDU frontier of which I am the first author. The corresponding material was largely written by me prior to any co-author contributions to the survey. The adaptation in this review trims, re-tones, and reorients the survey material toward the methodology research direction motivated in §§8–??; the remaining sections (Introduction, Background, Open Problems, Targets and Loci, and Conclusion) are original to this document.

Supporting Materials. The following supporting artefacts were produced during the preparation of this review:

- Paper inventory and cross-reference spreadsheet (surveyed systems, dataset and benchmark coverage, evaluation metrics): <https://docs.google.com/spreadsheets/d/1ZoPeIvnZKkHOQN8G3Fjj9gY8jnLImZvcYQrL36nj/edit?usp=sharing>
- Project repository (literature review $\text{\LaTeX}$ source, build script, bibliography, and supporting code): <https://github.com/LeweiXu/MP-VRDU/>
- Overleaf project for the companion survey paper: *URL pending*.

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